

Adversarial Vulnerabilities in Deep Image Classifiers: A Study on ResNet-34 and DenseNet-121

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Abstract

We evaluate adversarial attacks on ResNet-34 using FGSM, PGD, and patch methods. PGD significantly degraded accuracy, while patch attacks were less effective. Transferability to DenseNet-121 revealed shared vulnerabilities. Results emphasize the threat of gradient-based attacks and the need for robust defenses in deep learning systems.

1. Introduction

Deep neural networks (DNNs), while achieving state-of-the-art performance in image classification, are notoriously vulnerable to adversarial attacks—carefully crafted perturbations that cause models to misclassify inputs. This project investigates the susceptibility of a pre-trained ResNet-34 model to three types of adversarial attacks: Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and patch-based attacks. Additionally, we evaluate the transferability of these attacks to a DenseNet-121 architecture.

The experiments were conducted on a curated subset of the ImageNet-1K dataset, comprising 500 images across 100 classes. The baseline ResNet-34 model achieved 76.00% Top-1 accuracy and 94.20% Top-5 accuracy on unperturbed data. By introducing imperceptible perturbations bounded by L_∞ -norm constraints, we aimed to degrade model performance significantly. Our findings reveal that iterative attacks like PGD can reduce Top-1 accuracy to near-zero levels, while localized patch attacks are less effective despite larger perturbation budgets. Transferability experiments further demonstrate that adversarial examples designed for one model can partially compromise others, highlighting systemic vulnerabilities in DNNs.

The link to repository given here: <https://github.com/saltnpepper12/Project-3-Deep-Learning.git>

2. Methodology

The methodology for this project was designed to systematically evaluate the vulnerability of deep neural networks (DNNs) to adversarial attacks, focusing

on three primary strategies: Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and patch-based attacks. Each attack was tailored to exploit specific weaknesses in the ResNet-34 architecture, with subsequent analysis of their transferability to DenseNet-121. Below, we delve into the technical design, implementation nuances, and rationale behind hyperparameter choices.

2.1 Attack Design

Fast Gradient Sign Method (FGSM) The FGSM attack is a single-step gradient-based method that generates adversarial examples by perturbing input images in the direction of the loss gradient. Formally, for an input image x with true label y , the adversarial example x_{adv} is computed as:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, y))$$

where $\mathcal{L}$ is the cross-entropy loss, and ϵ controls the perturbation magnitude.

Implementation Details:

- **Perturbation Budget (ϵ):** Set to 0.02, corresponding to a maximum pixel change of ± 1 in the 0–255 range. This value balances imperceptibility and attack efficacy.
- **Normalization Handling:** Inputs were normalized using ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. Perturbations were clipped to ensure valid pixel ranges post-normalization.
- **Gradient Computation:** The gradient $\nabla_x \mathcal{L}$ was derived using PyTorch’s automatic differentiation, with gradients retained for the input tensor.

Rationale: FGSM was chosen for its simplicity and computational efficiency, serving as a baseline for comparing iterative methods. However, its single-step nature limits its ability to fine-tune perturbations for maximum damage.

Projected Gradient Descent (PGD) PGD extends FGSM by iteratively refining perturbations

within the L_∞ -norm constraint. Starting from a randomly initialized perturbation δ , the update rule at iteration t is:

$$\delta_{t+1} = \text{Clip}_\epsilon(\delta_t + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(x + \delta_t, y)))$$

where α is the step size, and Clip_ϵ projects perturbations back to the ϵ -ball.

Implementation Details:

- **Iterations:** 20 steps were chosen empirically to balance convergence and computational cost.
- **Step Size (α):** Set to $\epsilon/10 = 0.002$, ensuring gradual ascent without overshooting the ϵ -ball.
- **Random Initialization:** Perturbations were initialized uniformly in $[-\epsilon, \epsilon]$ to avoid local minima.
- **Clip Function:** Ensured $\|x_{\text{adv}} - x\|_\infty \leq \epsilon$ after each iteration.

Rationale: PGD was selected to test whether iterative optimization could overcome FGSM’s limitations. The multi-step process allows the attack to explore the loss landscape more thoroughly, maximizing misclassification.

Patch Attack Patch attacks restrict perturbations to a localized region (32×32 pixels) to simulate real-world adversarial stickers. The adversarial patch P is optimized to minimize the probability of the true class:

$$\min_P \mathcal{L}(x + P, y_{\text{target}})$$

where y_{target} is a least-likely class predicted by the model.

Implementation Details:

- **Patch Size:** 32×32 pixels, chosen to balance attack surface and perceptibility.
- **Perturbation Budget (ϵ):** Increased to 0.5 to compensate for spatial constraints.
- **Placement:** Patches were randomly positioned on the image to avoid bias.
- **Optimization:** 50 iterations of gradient ascent were used, with gradients computed only for the patch region.

Rationale: Patch attacks were implemented to evaluate spatial robustness. Larger ϵ values were necessary to offset the limited attack surface, reflecting the trade-off between perturbation size and spatial localization.

2.2 Hyperparameter Selection and Trade-offs

FGSM vs. PGD

- **FGSM:**
 - Pros: Fast (8.61 iterations/second) and memory-efficient.
 - Cons: Suboptimal due to single-step noise injection.

- **PGD:**

- Pros: Achieved near-total model collapse (0.20% Top-1 accuracy).
- Cons: $3.5\times$ slower than FGSM (2.98 seconds per batch).

2.3 Patch Attack Constraints

- **Spatial Limitations:** Localized perturbations left critical features outside the patch intact, limiting efficacy.
- **Budget ($\epsilon = 0.5$):** Larger than FGSM/PGD to counteract smaller attack surface.

Transferability to DenseNet-121

- **Shared Vulnerabilities:** FGSM and PGD perturbations transferred better due to architectural similarities (e.g., convolutional layers).
- **Patch-Specific Weaknesses:** DenseNet’s dense connections diluted localized perturbations, reducing transferability.

3. Results and Conclusion

3.1 Accuracy Metrics

The adversarial attacks implemented in this project produced stark differences in classification accuracy across the ResNet-34 and DenseNet-121 models. Below, we dissect these results to understand the mechanisms driving performance degradation and the implications of each attack type.

Baseline Performance (Original Test Set): The ResNet-34 model achieved 76.00% Top-1 accuracy and 94.20% Top-5 accuracy on unperturbed data, establishing a strong baseline. This performance aligns with expectations for a pre-trained model on ImageNet-1K, where Top-5 accuracy is typically higher due to the hierarchical nature of class labels (e.g., “German Shepherd” and “Golden Retriever” both being subsets of “dog”).

The gap between Top-1 and Top-5 metrics (18 percentage points) reflects the model’s ability to recognize broader categories even when uncertain about fine-grained details.

FGSM Attack: Rapid but Suboptimal Degradation With an L_∞ -norm constraint of $\epsilon = 0.02$, the FGSM attack reduced Top-1 accuracy to 11.40% and Top-5 to 37.60%. This represents a 64.60% absolute drop in Top-1 accuracy, demonstrating the vulnerability of ResNet-34 to single-step gradient-based perturbations.

The relatively higher Top-5 accuracy indicates that adversarial examples often remain within semantically related categories. For instance, an image perturbed to misclassify a “barbell” as a “CD player” might still retain features that place it in the “man-made object” supercategory. The imperceptibility of FGSM perturbations ($L_\infty \leq 0.02$) underscores how minimal pixel-level changes can exploit high-frequency patterns in the model’s decision boundaries.

PGD Attack: Near-Total Model Collapse The iterative PGD attack, executed over 20 steps with a step size $\alpha = 0.002$, achieved catastrophic results: 0.20% Top-1 and 10.60% Top-5 accuracy. This near-zero Top-1 accuracy highlights the power of multi-step optimization to refine adversarial perturbations.

Unlike FGSM, which applies a single coarse adjustment, PGD incrementally “pushes” the input across the model’s loss landscape, maximizing misclassification. The marginal Top-5 performance suggests that even iterative attacks struggle to completely obscure high-level semantic features (e.g., a perturbed “aircraft carrier” might still be recognized as a “vehicle” but misclassified as a “submarine”).

The success of PGD underscores the non-robustness of DNNs to carefully crafted adversarial trajectories.

Patch Attack: Limited Impact Despite Larger Budget Despite a significantly larger perturbation budget ($\epsilon = 0.5$), the 32×32 patch attack only reduced Top-1 accuracy to 59.60% and Top-5 to 84.00%. This modest degradation—16.40% absolute drop in Top-1 accuracy—reveals the challenges of localized attacks.

While the patch introduced visible artifacts (e.g., a distorted region in the corner of an image), critical features outside the perturbed area often remained intact, allowing the model to retain partial correctness. For example, a “bobsled” image with a patch might still exhibit recognizable contours of a sled, enabling correct classification.

The higher residual accuracy compared to FGSM and PGD suggests that spatially constrained attacks require more strategic placement (e.g., occluding object-specific features like wheels or faces) to maximize impact.

Transferability to DenseNet-121 Attacks designed for ResNet-34 exhibited varying degrees of transferability to DenseNet-121:

- **FGSM:** Top-1 accuracy dropped from 74.80% (clean) to 63.00%, a 11.80% reduction.
- **PGD:** Top-1 fell to 61.00%, a 13.80% reduction.
- **Patch:** Minimal impact, with Top-1 at 73.20% (1.60% drop).

The partial transferability of FGSM and PGD perturbations can be attributed to shared architectural features between ResNet and DenseNet, such as convolutional layers and skip connections, which learn similar high-frequency patterns.

However, the weaker transfer of patch attacks suggests that localized perturbations are more model-specific, as they exploit spatial dependencies unique to ResNet’s feature maps. The resilience of DenseNet-121—particularly to patch attacks—may stem from its densely connected layers, which aggregate features across multiple scales, diluting the impact of localized noise.

3.2 Key Observations:

1. Attack Potency vs. Computational Cost:

- FGSM’s single-step design enabled rapid generation (8.61 iterations/second) but achieved suboptimal results.
- PGD’s iterative process (2.98 seconds per batch) delivered superior efficacy at the cost of $3.5\times$ longer runtime.
- Patch attacks (7.62 seconds per batch) were computationally intensive yet less effective, highlighting the trade-off between spatial constraints and attack success.

2. **Semantic Preservation in Top-5 Results:** Even under strong attacks like PGD, Top-5 accuracy remained above 10%, indicating that adversarial perturbations rarely erase all semantic information. This aligns with the hypothesis that DNNs rely on distributed feature representations, making complete erasure of class-specific signals difficult.

3. **Transferability as a Double-Edged Sword:** The transferability of FGSM and PGD attacks to DenseNet-121 implies that adversarial examples exploit universal vulnerabilities in convolutional networks. This has dire implications for real-world systems, where attackers could use surrogate models to craft perturbations that compromise black-box targets.

3.4 Qualitative Analysis

The adversarial examples generated by FGSM, PGD, and patch attacks exhibited distinct visual characteristics that correlated with their effectiveness in fooling the ResNet-34 model. Below, we dissect these patterns and their implications for model vulnerabilities.

FGSM: Subtle Noise, Global Disruption FGSM perturbations introduced high-frequency, low-magnitude noise across the entire image, creating imperceptible changes to the human eye but significant distortions to the model’s feature maps.



Figure 1: Task 2: FGSM Attack Visualization

For example: These perturbations exploited the model’s sensitivity to high-frequency gradients in early convolutional layers. While humans perceive the image as unchanged, the noise altered activations in shallow filters, propagating errors through deeper layers.

PGD: Structured Adversarial Textures PGD’s iterative process generated targeted adversarial textures that systematically distorted key features. Unlike FGSM’s uniform noise, PGD perturbations resembled abstract patterns superimposed on the original image:



Figure 2: Task 3: PGD Attack Visualization

- The “CD player” image developed checkerboard-like distortions on its buttons and display, causing a misclassification to “cassette player.” The iterative refinement allowed PGD to amplify high-confidence adversarial features, such as edges resembling cassette tape slots.

PGD’s perturbations often resembled class-specific adversarial triggers. For instance, perturbations on animal images introduced stripe-like patterns resembling fur textures, while vehicles developed wheel-like circular artifacts. This suggests PGD leverages hierarchical feature dependencies in deeper layers to “hijack” class predictions.

Patch Attacks: Localized but Conspicuous Distortions The 32×32 patch attacks introduced visibly jarring artifacts in random regions of the image. Despite their conspicuous nature, these perturbations were less effective due to their spatial constraints:



Figure 3: Task 4: Patch Attack Visualization

- In the “bobsled” example (Task 4 visualization), a patch placed on the sled’s front introduced bright, chaotic pixels. While the model incorrectly classified it as “cassette,” the sled’s overall shape and icy background remained intact, preserving partial correctness.

Patch efficacy depended heavily on occlusion of critical features. Successful attacks required patches to cover object-specific regions (e.g., a dog’s face or a car’s wheels). Random placement, as implemented here, often left enough intact features for the model to infer the correct class.

4. Discussion and Implications

The results underscore the fragility of DNNs to gradient-based attacks. PGD’s near-total success highlights the need for robust defenses such as adversarial training, where models are trained on perturbed examples to improve resilience. Patch attacks, while less

effective, pose real-world risks; for instance, adversarial stickers on traffic signs could mislead autonomous vehicles.

The partial transferability of attacks to DenseNet-121 suggests that adversarial examples exploit universal features learned by convolutional networks. This has critical implications for security-critical applications, as attackers might not need full knowledge of the target model’s architecture.

5. Conclusion

This project systematically evaluated adversarial attacks on ResNet-34, revealing profound vulnerabilities to gradient-based methods like PGD. Key takeaways include:

1. **Iterative Attacks Dominate:** PGD’s multi-step optimization achieved near-100% success, far surpassing FGSM.
2. **Localized Perturbations Are Challenging:** Patch attacks required larger budgets but were less effective, highlighting the difficulty of spatial constraints.
3. **Transferability Exists but Varies:** Attacks generalized moderately to DenseNet-121, suggesting shared vulnerabilities across architectures.

Future work could explore hybrid attacks combining spatial and gradient-based perturbations or investigate defense mechanisms like adversarial training. As DNNs proliferate in security-critical systems, understanding and mitigating adversarial risks remains paramount.

References

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